Comparative Analysis of Deep Learning and Traditional Approaches for 3D Reconstruction

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***Abstract*— Advances in 3D reconstruction techniques have led to significant progress in various areas, including virtual reality, augmented reality, robotics, and more. However, their potential impact on medical imaging and diagnostic applications is also noteworthy. This article provides a detailed exploration of both traditional and modern methods in 3D reconstruction, aiming to explain their strengths, limitations, and potential impact on different fields. The article discusses the technical details of traditional SFM methods and modern NeRF-based approaches, comparing them to understand how they are evolving. By analyzing data from academic papers and tools like COLMAP and NerfStudio, the study offers insights into how effective and reliable these reconstruction techniques are. Experimental results show that NeRF-based methods, such as the Nerfacto model, perform better than traditional SFM and Multi-View Stereo (MVS) techniques in creating detailed point clouds. This superiority is particularly promising in industries like healthcare, where detailed 3D reconstructions are crucial for medical imaging and diagnostics. The research was conducted as part of the project APVV-21-0502: BrainWatch.**

***Keywords*—3D Reconstruction; Structure From Motion; Neural Networks; Point Clouds; COLMAP; NerfStudio**

I. Introduction

Creating accurate 3D reconstructions from 2D images is a fundamental task in computer vision, with wide-ranging applications in fields such as virtual reality, augmented reality, and robotics. Over the years, various techniques have been developed to tackle this challenge, ranging from traditional image processing methods to advanced algorithms utilizing deep learning and neural networks [1], [2].

Traditional methods rely on extracting features from images, estimating camera poses, and triangulating points to reconstruct a 3D scene incrementally. These methods have been widely used and studied, offering a robust framework for 3D reconstruction. However, they often suffer from limitations in accuracy, and efficiency, especially when dealing with complex geometric structures. On the other hand, Nerf-based techniques, inspired by Neural Radiance Fields (Nerf [3]), represent a paradigm shift in 3D reconstruction. These methods leverage neural networks to directly learn the mapping from 2D images to 3D scene representations, bypassing the intermediate step of feature extraction and pose estimation. Nerf-based approaches offer the potential for highly detailed and photorealistic reconstructions by modeling the entire scene as a continuous volumetric function [3], [4].

In recent years, the emergence of Neural Radiance Fields (NeRF) and related methods has revolutionized the field of 3D reconstruction. NeRF-based techniques, such as Neural Scene Representation, leverage deep neural networks to directly model scene geometry and appearance from 2D images. By learning implicit scene representations, these methods offer the potential for generating highly detailed, photorealistic reconstructions with unprecedented fidelity. NeRF-based approaches operate by training neural networks to approximate the volumetric density and color of 3D scenes based on input images and corresponding viewing parameters. This enables the generation of novel views of the scene from arbitrary viewpoints, facilitating applications such as immersive virtual reality, digital content creation, and synthetic image generation. Furthermore, advancements in optimization algorithms, such as differentiable rendering and implicit function fitting, have improved the efficiency and scalability of NeRF-based techniques. Additionally, the integration of neural networks with traditional geometric methods, such as SfM and MVS, has led to hybrid approaches that combine the strengths of both paradigms for enhanced reconstruction quality and efficiency [5], [6].

II. Materials and Methods

The important relationship between Structure From Motion (SFM) and Multi-View Stereo (MVS) techniques was understood, so efforts were made to combine them for a more thorough reconstruction process. SFM, known for its ability to turn 2D image sequences into detailed 3D structures, formed the basis of our approach. Meanwhile, MVS techniques, seamlessly integrated into tools like COLMAP, allowed us to capture intricate details from various viewpoints, enhancing the accuracy of our reconstructions. Expanding our toolkit, COLMAP [7] and NerfStudio [8] were incorporated, both powerful applications with distinct advantages in 3D reconstruction. COLMAP's fusion of SFM and MVS techniques offered a comprehensive approach, capturing details from multiple angles and ensuring thorough evaluations. On the other hand, NerfStudio utilized neural networks to delve into the finer details of our subject matter, resulting in precise and faithful reconstructions. For refining and aligning point clouds extracted from datasets the Blender [9] was used. Its flexibility allowed us to carefully prepare each point cloud (PC).

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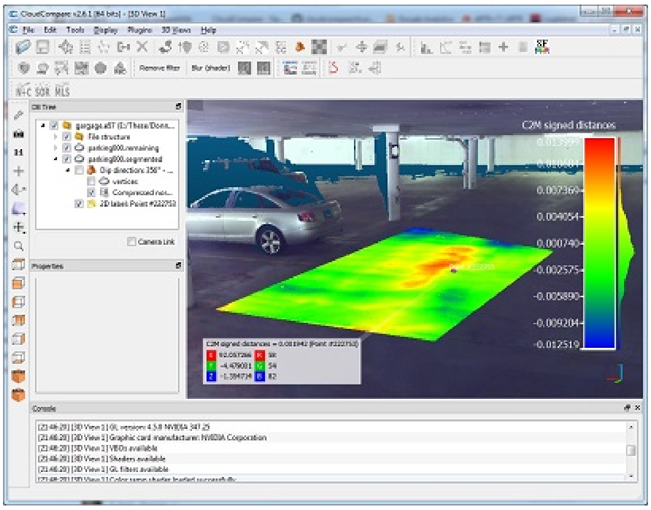

Figure 1. CloudCompare

As accurate alignment was crucial, CloudCompare (see Fig. 1) emerged as our trusted companion. By using its registration methods, CloudCompare enabled us to align datasets perfectly, reducing discrepancies and laying the foundation for fair comparisons. One notable technique employed within CloudCompare [10] is the Hoffman algorithm, a powerful tool for surface reconstruction. This algorithm works by iterative fitting surfaces to the point cloud data, refining the fit with each iteration until convergence is achieved. By leveraging the Hoffman algorithm, detailed surfaces were reconstructed from sparse point cloud data, enriching the fidelity of the reconstructions.

The method of squares within CloudCompare was utilized for comparing point clouds, involving calculating the distances between corresponding points in two point clouds and squaring these distances to eliminate negative values. The mean square distance is then computed to quantify the difference between the point clouds, aiding in assessing reconstruction accuracy by providing a quantitative measure of dissimilarity.

Using Iterative Closest Point (ICP) registration, at least four common points were selected on each point cloud, allowing the software to align, rotate, and scale the point clouds accordingly.

This iterative process minimized any discrepancies between the point clouds, resulting in a very small error rate. In addition to the methodology outlined earlier, the nearest neighborhood (KNN) (see Fig. 2) method was employed for comparing point clouds. Utilizing KNN allowed us to efficiently identify corresponding points between the datasets.

III. Experimental Results

In the quest to understand the complexities of 3D reconstruction, a range of tools and methods was carefully selected, each playing a crucial role in the efforts. The endeavor began with establishing a robust hardware foundation, specifically designed to support the computational tasks at hand. With a powerful Windows PC equipped with an NVIDIA 3060 video card, the calculations were sped up using GPU processing, ensuring that analyses were quicker and more efficient. This setup provided us with the computing power needed to tackle the challenges of 3D reconstruction. For thorough testing and validation of methods, the Anaconda environment and virtual environment (venv) with Python 3.11 were utilized, ensuring experiments were reproducible and consistent across the board. Each tool and method chosen played a specific role in our overarching goal of exploring of 3D reconstruction, from robust applications to meticulous methodologies, and from hardware infrastructure to software configurations.

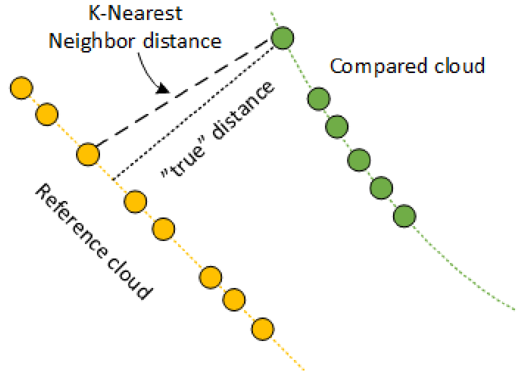

Figure 2. KNN-Based point cloud comparative analysis

The research aims to evaluate and compare various methods of 3D reconstruction, specifically focusing on comparing traditional Structure From Motion (SFM) techniques with modern approaches integrating neural networks. Experiments were conducted using point cloud data obtained from two academic papers detailing SFM-based reconstruction as primary reference sources. Additionally, results obtained from the NeRF method using data from COLMAP were compared.

Figure 3. Dataset "Gustav II Adolf" [12]

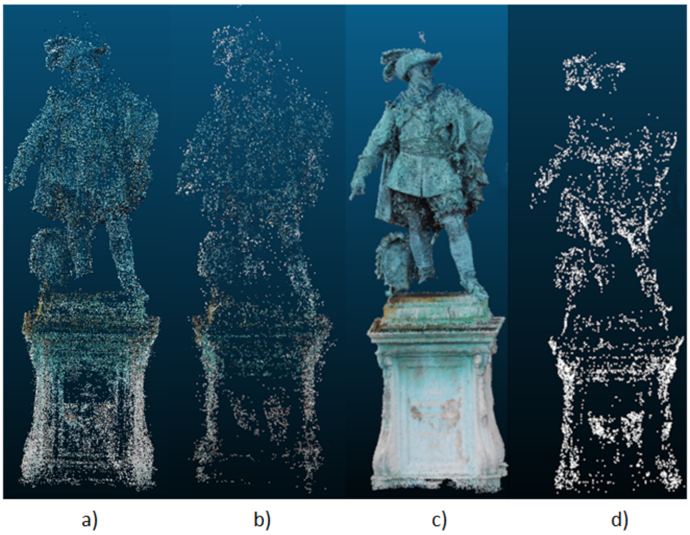

Figure 4. Comparative analysis of point clouds: a) COLMAP, b) Yasha KS [13], c) Nerfacto, d) Carl Olsson [14]

To compare the reconstructed point clouds, the CloudCompare application, widely used for point cloud processing and analysis, was employed. Our comparison focused on evaluating the distances between points in the reference point cloud obtained from each method. A standardized dataset [11] (see Fig. 3) named "GustavIIAdolf" was utilized, consisting of 57 photos of the same object from various angles and calibration matrix [12].

The point cloud data from two primary sources and COLMAP and NerfStudio applications was acquired (see Fig. 4). For comparison, datasets published in [13], [14] were used. Python scripts were utilized to implement the reconstruction process described in the paper, generating a point cloud.

Our next step was to align and refine of point clouds for comparison. To accomplish this, Blender, a powerful 3D modeling and animation software, was used. Each point cloud was visually inspected, identifying and removing any irrelevant points that could skew the comparison. To address the removal of irrelevant points, we utilized a combination of manual inspection and automated tools provided by NerfStudio. While some points were removed manually based on visual inspection to ensure accuracy and relevance, we also employed a method within NerfStudio that allows for the selection of points through the use of a bounding box. By adjusting this bounding box, we could isolate and retain only the points that accurately represented the reconstructed scene. This ensured that the relevant data points, allowing for a more accurate assessment of the reconstruction methods were compared. After refining the point clouds to a common reference point were aligned. Each point cloud was positioned at the origin point (0, 0, 0), and their sizes were uniformly adjusted to ensure consistency across all four datasets. This step was crucial to ensure that the comparison was fair and unbiased, as it eliminated any variations in position and scale between the different reconstructions. Once the alignment and refinement process was completed, four point clouds were positioned identically, with similar sizes and orientations. Next, the point clouds were imported into the CloudCompare for further processing. While the alignment and sizing done in Blender provided a good starting point, ensuring precise alignment was crucial for an accurate comparison. In CloudCompare, the method known as Iterative Closest Point (ICP) registration was employed to achieve precise alignment. Through meticulous alignment of the point clouds in CloudCompare, the comparative analysis was ensured to be based on accurate and consistent data. With the point clouds aligned and standardized, a series of comparisons between different reconstruction methods were conducted.

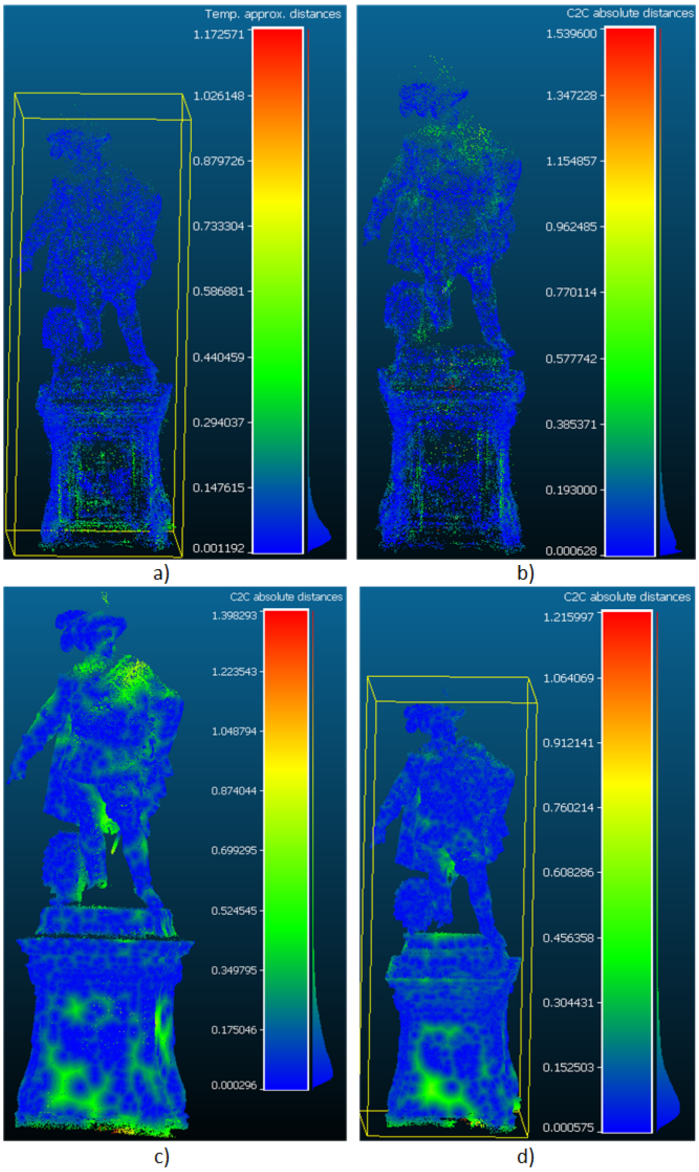

Figure 5. Comparative evaluation: a) Yashas KS SFM (reference) vs. COLMAP point cloud (comparison), b) Carl Olsson SFM (reference) vs. COLMAP point cloud, c) Carl Olsson SFM (reference) vs. Nerfacto model, d) Yashas KS SFM (reference) vs. Nerfacto model

Firstly, the SFM method proposed by Yashas KS with the COLMAP point cloud, which integrates SFM and MVS techniques, for generating 3D reconstructions was compared (see Fig. 5). This comparison provided insights into the effectiveness and reliability of traditional SFM techniques in 3D reconstruction. Afterward, the comparison was repeated using the SFM method proposed by Carl Olsson with the COLMAP point cloud (see Fig. 5). Next, the point cloud produced by Yashas's SFM method was compared with one generated by the Nerfacto model (see Fig. 5). Nerfacto, a neural network-based approach, promised to deliver highly detailed reconstructions by leveraging the same image dataset used for generating the other three point clouds. As a final step, the comparison was repeated by evaluating the point cloud produced by Carl Olsson's SFM method with the Nerfacto model (see Fig. 5). Among the methods compared, Nerfacto stood out for its ability to produce the most detailed PC.

TABLE I. Point Count Analysis of Generated Point Clouds

	SFM method by Yashas [13]	**SFM method by Carl Olsson [14]**	**SFM and MVS using COLMAP [10]**	**Nerf based method using Nerfacto [11]**
Count of points	18528	5805	59640	1000398

IV. Conclusion

In conclusion, our comparative analysis highlights the superior performance of the Nerfacto model in generating more detailed point clouds. This is evidenced by the point counts presented in the Table 1, where Nerfacto consistently produces higher counts compared to COLMAP and other methods. Furthermore, our examination of the point clouds through CloudCompare reveals that Nerfacto results in fewer gaps within the reconstructed model when compared to COLMAP. These findings underscore the efficacy of the Nerfacto model in producing highly detailed and comprehensive 3D reconstructions, demonstrating its potential for medical imaging, diagnostic applications, or scientific research.

The methods used in this article were selected to ensure a comprehensive comparison between traditional and modern 3D reconstruction techniques. NeRF-based methods have shown significant promise in generating highly detailed and photorealistic 3D representations by learning the volumetric scene function directly from image inputs. Despite NeRF not inherently producing point clouds as an end product, its ability to reconstruct intricate details with higher fidelity makes it a compelling candidate for this comparison. It's also important to note that SfM and MVS methods can be utilized independently for comprehensive 3D reconstruction tasks. Tools such as COLMAP integrate both SfM and MVS to improve the quality and detail of reconstructions. When comparing traditional SfM/MVS methods with NeRF-based approaches, their capabilities were assessed both as standalone techniques and as components within more complex pipelines.

Future research could explore comparisons with other NeRF-based methods like INR-NeRF, Instruct-GS2GS, Splatfacto, etc., to deepen understanding of their strengths and weaknesses across different domains. Additionally, investigating NeRF methods for medical applications, particularly in brain reconstruction, could lead to better insights into brain anatomy and pathology, enhancing diagnostic accuracy and treatment outcomes in neurological disorders. Overall, the use of 3D reconstruction for aneurysm detection enhances diagnostic accuracy, facilitates surgical planning, and improves patient outcomes by providing clinicians with detailed insights into the vascular anatomy.

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